A Stable Disk Collapses Without Warning

TESS catches a nova-like disk crumpling while an odd 45-day eruption unfolds.

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Background

Nova-like variables accrete so fast that standard disk instability theory predicts thermally steady, outburst-free disks. Their occasional flares and fades are usually blamed on varying mass transfer from the donor.

> **No clear mechanism explains flaring in thermally stable disks without invoking mass transfer changes.**

What this paper is selling

A fading episode reveals a thermally stable disk dynamically collapsing, tracked by a negative superhump drifting toward the orbital frequency. The simultaneous eruption and collapse pull mass transfer in opposite directions, demanding a new dynamics-driven cycle.

Key results to remember

- Negative superhump emerged and evolved toward the orbital frequency during the collapse.
- A symmetric 45-day eruption coincided with the collapse, contradicting mass-transfer expectations.
- Disk cycled circular, eccentric, tilted, then minimal over about 700 days.

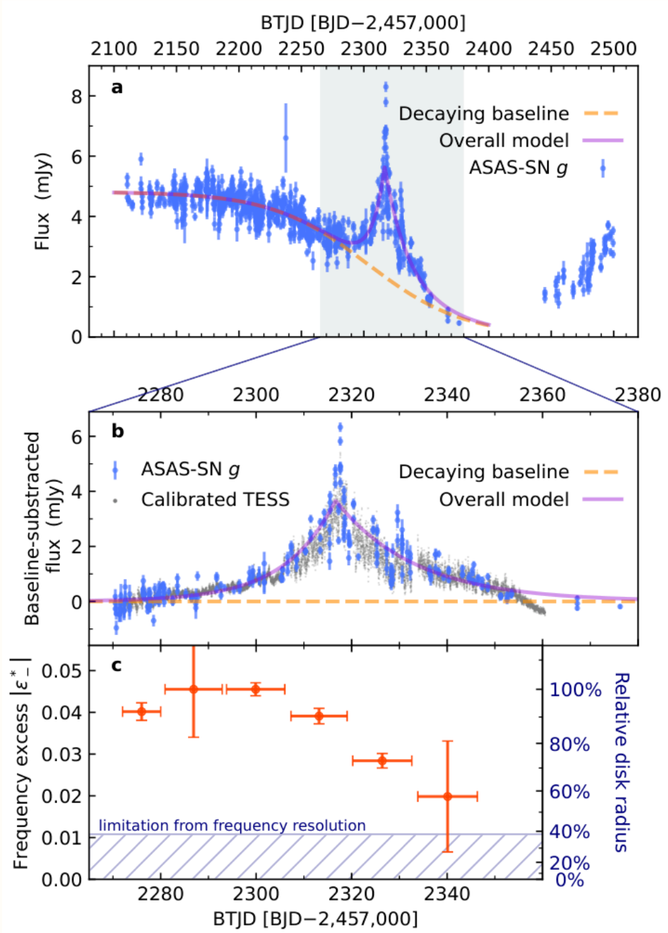


Figure 3 | The negative superhump frequency climbs toward the orbital frequency exactly as the 45-day eruption unfolds.

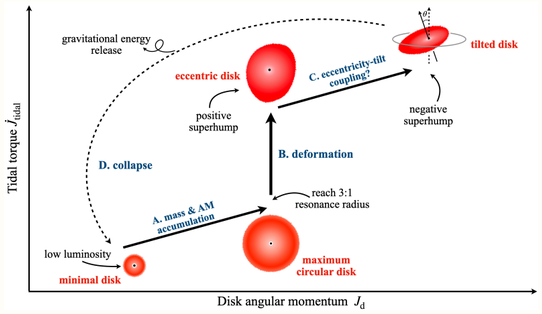


Figure 5 | A proposed cycle: the disk grows, turns eccentric, tilts, then catastrophically collapses back to a minimal state.